- Jordan imports most of its energy and seeks greater energy security as well as lower electricity prices.
- The country was aiming to have two 1000 MWe nuclear power units in operation by 2025, but is now considering the use of small modular reactors instead.
- Jordan has significant uranium resources, some in phosphorite deposits.

Privacy - Terms

Electricity sector

Total generation (in 2023): 23.8 TWh

Generation mix: natural gas 14.7 TWh (62%); solar 3.6 TWh (15%); coal 3.0 TWh (13%); wind 1.6 TWh (7%); oil 0.9 TWh (4%).

Import/export balance: 0.1 TWh net import (0.3 TWh imports, 0.2 TWh exports)

Total consumption: 21.3 TWh

Per capita consumption: c. 1900 kWh in 2023

Source: International Energy Agency and The World Bank. Data for year 2023.

Nuclear power industry

Jordan imports over 95% of its energy needs, at a cost of about one-fifth of its GDP.

It has about 5200 MWe of generating capacity and electricity consumption is growing at about 3% a year. Jordan has regional grid connections of 500 MWe with Egypt, 300 MWe with Syria, and it is increasing links with Israel and Palestine.

Also it has a 'water deficit' of about 600 million cubic metres per year (1500 m^3 demand, 900 m^3 supply). It pumps over 60 million m^3/yr of fossil subartesian water from the Disi/Saq aquifer, which contains elevated, but not hazardous, levels of radionuclides, principally radium. (Drinking 2 litres per day would give a dose of 1.0 to 1.5 mSv/yr.)

Nuclear power plans: large reactors

Jordan's Committee for Nuclear Strategy, set up in 2007, set out a program for nuclear power to provide 30% of electricity by 2030, and to provide for exports. The nuclear law was modified in 2007 to

establish the Jordon Atomic Energy Commission (JAEC) and the Jordan Nuclear Regulatory Commission (JNRC), including radiation protection and environmental roles.

JAEC's functions include safety and security, nuclear science and technology, and safeguards and verification. Its commission is to transform Jordan from net energy importer to net electricity exporter by 2030, to provide power to fuel economic growth at low cost, and to end dependence on fossil fuels. Its strategy includes exploiting national uranium assets, promoting public/private partnerships, ensuring effective technology transfer and national participation, providing for water desalination and eventually hydrogen production, developing spin-off industries, and enabling competitive energy-intensive industries.

In mid-2008 an agreement between JAEC and Atomic Energy of Canada Ltd (AECL) with SNC-Lavalin was to conduct a three-year feasibility study on building an AECL 740 MWe Enhanced Candu-6 reactor using natural uranium fuel, for power and desalination. In August 2008 it was reported that the government intended to sign up for an Areva (now Framatome) reactor, and subsequent discussions pointed to an 1100 MWe Atmea unit.

In December 2008 JAEC signed a memorandum of understanding with Korea Electric Power Corp (KEPCO, parent company of KHNP) to carry out site selection and feasibility study on nuclear power and desalination projects. This was related to Doosan Heavy Industries, Korea's main nuclear equipment maker, carrying out desalination-related work in Jordan under a separate recent agreement, and KEPCO having won a tender to build a 400 MWe gas-fired power plant on a build-own-operate basis. Up to 40% of the capacity of any nuclear plant built on the coast would likely be used for desalination.

Site options with seawater cooling are limited to 30 kilometres of Red Sea coast near Aqaba. In September 2009 JAEC contracted with Tractabel Engineering, a subsidiary of GdF Suez, to undertake a two-year siting study for a new plant some 25 km south of Al Aqabah and 12 km east of the Gulf of Aqaba coastline. Discussion of environmental aspects took place with Israel and Egypt. However, late in 2010 the proposed location for the first reactor became Al Amra in the Majdal area in northern Al Mafraq province, about 70 km from Amman, due to better seismic characteristics, and Tractabel turned its attention to defining a site there. The plant needs to have PGA seismic level of 400 gal for safe shutdown. Cooling water would come from the municipal Khirbet Samra Wastewater Treatment Plant, with the cooling system modelled on

that at Palo Verde in Arizona, USA, which also uses wastewater for cooling.

The site then changed to Qasr-Amra in Al-Azraq province, about 70km south east of Amman. In April 2014 an expert team from the International Atomic Energy Agency (IAEA) visited the region to evaluate studies on the proposed site, which is some 70 km from the cooling water source at Samra, and well away (100 km) from a fault line.

Earlier in November 2009 JAEC signed an $11.3 million agreement with WorleyParsons for the pre-construction phase of a nuclear power plant with two 1000 MWe-class reactors. The firm carried out technology selection – preparing the tender and evaluating bidders – as well as assisting in fuel cycle engineering and waste management plans for the plant.

Also in 2009 the JAEC evaluated seven offers from at least four reactor vendors, and in May 2010 three vendors and designs were shortlisted – the Atmea1 from Areva-MHI, the AECL EC6, and the AES-92 from Atomstroyexport (ASE), Rosatom's reactor export subsidiary. In April 2012 the field was narrowed to two: Atmea1 and AES-92, both around 1000 MWe. JAEC and WorleyParsons pursued discussions first on the actual reactor designs and secondly "on the financing and organization support that the vendor will be providing for future operation of the plant." For Jordan, this is the major consideration.

However, in May 2012 the lower house of parliament voted 36 to 27 in favour of a recommendation by the parliamentary Energy & Mineral Resources Committee to suspend the country's nuclear program, including uranium exploration. The committee's report accused the JAEC of misleading parliament. However, JAEC said the motion was qualified in effect to endorse its caution in proceeding, and developments since then support this.

In October 2013 JAEC announced that ASE would be the supplier of two AES-92 nuclear units, while Rusatom Overseas would be strategic partner and effectively the operator of the plant through a joint venture. It was expected that Rusatom Overseas would contribute 49.9% of the project's $10 billion cost, with the state-owned Jordan Nuclear Power Co (JNPC) being responsible for the controlling 50.1%. In September 2015 JAEC said that it was negotiating for the Industrial and Commercial Bank of China with China National Nuclear Corporation (CNNC) to finance not less than 50% of the construction project, though the main nuclear island would use Russian technology. The plant was to be provided on a

build-own-operate (BOO) basis,* but JAEC suggested that a final split of share capital in the plant might be Jordan 35%, Russia 35% and China 30%. Rosatom was to supply all the fuel and take back the used fuel.

* In May 2013 JAEC said that the two proposals had been too expensive and no strategic investor had been found, so it was now considering reopening bids to include small modular reactors which would better suit its overall electricity supply situation. This would be logical given the small grid, but for an initial plant the idea appears to be moot.

In September 2014 JAEC signed a project development agreement with Rusatom Overseas**, with a view to a final construction contract within two years. An intergovernmental agreement was signed in March 2015, outlining responsibilities for stage 1 of the project, including setting up the JNPC project company. An intergovernmental pre-investment agreement was signed in September 2015. In September 2016 Rosatom said that a feasibility study for the $10 billion project would be completed in 2017. In June 2018 Jordan announced that it had scrapped the agreement. JAEC stated that it was deemed too costly as a result of Rosatom wanting to secure finance through commercial loans. The two units would have provided about half of Jordan's electricity and enabled exports to Syria and Iraq. JAEC stated that it would explore the use of small modular reactors (SMRs). Rosatom said it hoped the proposal would be revived at some point in the future.

** Earlier, in February 2011, the Energy Minister announced that JAEC had shortlisted GDF Suez, Rosatom, Datang International Power Generation Co. and Kansai Electric Power Co. as possible strategic partners to invest in and operate the new plant of about 2000 MWe, with the government retaining 26-51% of the equity. Jordan was looking at limited recourse financing with a debt-equity ratio of at least 70-30, and wanted to set up a long-term (e.g. 45-year) power purchase contract, with the government guaranteeing part of the debt. The overnight cost was expected to be $4900/kW, hence likely $9.8 billion apart from financing, though a later estimate was €12 billion. JAEC expected to start building one 750-1200 MWe nuclear power unit for operation by 2020 and a second one for operation by 2025. The UAE was reported to be supporting the project and exploring collaborative arrangements.

At Jordan's invitation, in August 2014 an IAEA integrated nuclear infrastructure review (INIR) mission reviewed the country's preparations for nuclear power and reported favourably. A phase 1 INIR mission had been in 2009, with follow-up in January 2012. A phase 2 INIR mission in 2015 reported in August that improvement

was needed in infrastructure development, regulation, and coordinated government activity. In November 2015 an international advisory group was set up by the government to review progress in implementing the nuclear energy programme.

In November 2018, JAEC Chairman Khaled Toukan told an IAEA ministerial conference that, although the country was working on two parallel projects of a single nuclear reactor in the 1000 MWe range and an SMR plant, the SMR project "seems to be the more appropriate in bridging the gap in the Jordanian electricity generation mix."

Nuclear power plans: small reactors

Small reactors are on JAEC's agenda, originally to follow the first large one(s). In November 2013 JAEC said it would build several small reactors of about 180 MWe capacity.

In March 2017 an agreement between JAEC and Saudi Arabia's King Abdullah City for Atomic and Renewable Energy (KA-CARE) was signed for a feasibility study on construction of two SMRs in Jordan for the production of electricity and desalinated water. No particular technology was mentioned, but KA-CARE has an agreement with Korea Atomic Energy Research Institute (KAERI) to build its 330 MWt (100 MWe) SMART pressurized water reactor, and as described above and in the Research and development section below, JAEC has close ties with KAERI and KEPCO.

In November 2017 JAEC signed a memorandum of understanding with Rolls-Royce to conduct a feasibility study for the construction of an SMR, and another with X-energy to consider building that company's 76 MWe Xe-100 high temperature gas-cooled reactor.

In April 2018 JAEC said it was in advanced negotiations with China National Nuclear Corporation (CNNC) to build a 220 MWe HTR-PM high temperature gas-cooled reactor for operation from 2025.

Following the cancellation of the two planned Russian VVER units in June 2018, Rosatom stated that it and JAEC had stepped up cooperation on SMRs.

By November 2018, JAEC had signed SMR cooperation agreements with CNNC, Rolls-Royce, NuScale, X-energy and Rosatom.

In October 2023 an IAEA expert mission concluded that reports compiled by the JAEC provided a good basis for a feasibility study

into the country's use of SMRs for electricity generation and desalination.

Uranium resources and conversion

The 2024 OECD NEA/IAEA 'Red Book' lists 63,019 tU as in situ identified conventional resources (recoverable below $260/kgU), these being surficial and amenable to open pit mining. The total includes 35,029 tU JORC-compliant in central Jordan, and an estimated 27,990 tU in the Hasa-Qatrana area. Some 100,000 tU is estimated in phosphate deposits as a possible by-product. There have been feasibility studies on recovering uranium as a by-product of phosphate production, and R&D on this is ongoing.

In March 2025 Jordan Uranium Mining Company (JUMCO) signed a memorandum of understanding with Kazatomprom to study uranium exploration and mining projects in Jordan in order to assess their potential for further development. This was followed by the signing of a letter of intent on the development of the Central Jordan Uranium Project in August 2025. JUMCO is the commercial arm of the Jordanian Atomic Energy Commission (JAEC) and was established in 2013 to carry out radioactive elements exploration and development in Jordan. Its exploration and development efforts are focused on the Central Jordan Uranium Project, where the pilot plant began operation in early 2021.

Central Jordan: In October 2008 a joint venture between JAEC and Areva (now Orano) was established to define uranium resources in central Jordan, and in February 2010 this became the JV company Nabatean Energy. Areva in 2010 secured an agreement giving it exclusive uranium mining rights in central Jordan for 25 years. Areva said its goal was "to create a full partnership with Jordan on training and obtaining nuclear technology".

About 2008 the Jordan French Uranium Mining Company (JFUMC) was set up as a joint venture between Areva and Jordan Energy Resources Inc. (JERI) and traded as Jordan Areva Resources. It operated within a 1400 sq km concession area in the central region, including the Siwaqa, Khan Azzabib, Wadi Maghar and Attarat areas. In June 2012 JFUMC said it had identified over 20,000 tU resources in a 72 sq km area and would carry out a feasibility study on mining. The Minister for Energy and Mineral Resources earlier said that development of an open pit mine would begin in 2013, for operation from 2015. However, in October 2012 JAEC terminated the JFUMC joint venture mining licence due to JFUMC failing "to

submit its report on time". Areva insisted that the actual JFUMC agreement related only to exploration, not mining, and expired naturally in 2012 anyway.

In January 2013 the state-owned Jordan Uranium Mining Co (JUMCO) was established with the same broad remit as JFUMC. In May 2014 JUMCO announced that it would build a 300-400 tU/yr uranium mill in the country's central region, about 80 km south of Amman, where resources of 33,300 tU at about 0.011%U are known. It reported that its pilot plant was operating from early 2021 at this Central Jordan Uranium Project (CJUP). In May 2022 JUMCO reported that it had produced 20 kg of U_3O_8 from 160 tonnes of uranium ore.

Earlier in March 2017 an agreement between JAEC and Saudi Arabia's King Abdullah City for Atomic and Renewable Energy (KA-CARE) was signed covering uranium exploration and mining of uranium in central Jordan. Then in March 2019 KA-CARE launched a programme in conjunction with JAEC and JUMCO to develop Saudi expertise in uranium exploration and mining.

Some 28,700 tU is reported at **Hasa and Qatrana**, 80 km south of Amman, following 2010-11 work by Jordan Energy Resources Inc (JERI), a commercial arm of JAEC. This is in the Qatrana phosphorites, where uranium at 0.015-0.017% would be a co-product with phosphates and vanadium. About 52 Mt of phosphate is reported, but neither this nor the uranium is as JORC-compliant resources. JERI called for bids from major mining companies to develop seven separate blocks comprising the deposit late in 2012.

Some uranium mineralization is also reported at Rweished near the Iraq border in the far northeast. China National Nuclear Corporation (CNNC) has been searching for uranium at Hamra-Hausha in the north, and Wadi Baheyya in the south, while Rio Tinto was searching in Wadi Sahab/Sahra Abiad, close to the Saudi Arabian border, but withdrew in 2011.

Conversion

In May 2022 Jordan Uranium Mining Company (JUMCO) announced the production of 20 kg of yellowcake from 160 tonnes of uranium ore at a newly operational pilot processing facility. JUMCO is the commercial arm of the Jordanian Atomic Energy Commission (JAEC) and was established in 2013 to carry out radioactive elements exploration and development in Jordan.

Research and development

In December 2009 the JAEC selected a consortium headed by KAERI with Daewoo to build the 5 MWtJordan Research and Training Reactor (JRTR) at the Jordan University for Science and Technology at Al Ramtha – the country's first. Construction started in August 2013 following JNRC final approval, and the cost was expected to be $173 million. The reactor, similar to South Korea's 30 MW HANARO reactor, uses 19% enriched fuel and has the potential to be upgraded to 10 MWt. First criticality was in April 2016 and the reactor was commissioned in December. It was subject to an IAEA Integrated Nuclear Safety Assessment of Research Reactors (INSARR) peer review in December 2016.

In May 2017 JAEC signed an agreement with the French CEA to develop collaborative research programmes in connection with its International Centre based on Research Reactors (ICERR) research hub under a programme launched by the IAEA in 2014. The ICERR programme will contribute to enhancing the utilization of the JRTR and associated scientific equipment in Jordan, the CEA said.

The Korean consortium was reported to be bidding against firms from Argentina, China and Russia. It was financed partly by a $70 million soft loan from South Korea, with 0.2% interest rate and repayment over 30 years. The JRTR will serve as an integral part of the nuclear technology infrastructure and will become the focal point for a nuclear science and technology centre (NSTC) with a key role in educating and training future generations of nuclear engineers and scientists. It will supply radioisotopes for medicine, industry and agriculture.

JAEC also has the Jordan Subcritical Assembly operational at the same site, commissioned in 2013.

In May 2017 the new Synchrotron-light for Experimental Science and Applications in the Middle East (SESAME) was commissioned at Allan, 30 km northwest of Amman. This is the first synchrotron in the Middle East and will be used for advanced research projects by scientists from across the region**.** It is closely linked to the European Organization for Nuclear Research (CERN) and uses some

components from the decommissioned BESSY I synchrotron in Germany. It was developed under the auspices of UNESCO with support from the IAEA. It is funded by $6 million per year annual contributions from the joint venture's members of Cyprus, Egypt, Iran, Israel, Jordan, Pakistan, the Palestinian Authority and Turkey. Construction began in 2004 with capital provided by the governments of Jordan, Israel, and Turkey, and by the European Union (through CERN and directly) and Italy. Beamlines are being added progressively, with the third operational from December 2020.

Organization and regulation

In 2007 the nuclear law was amended to establish the Jordan Atomic Energy Commission (JAEC) and the Jordan Nuclear Regulatory Commission (JNRC), including radiation protection and environmental roles. In April 2014 JNRC was merged into the Energy and Minerals Regulatory Commission (EMRC).

In April 2015 the EMRC signed an agreement with the Korea Institute of Nuclear Safety (KINS) to help regulate Jordan's new research reactor.

The Ministry of Science, ICT and Future Planning (MSIP) is responsible for the JAEC.

International agreements

Jordan has signed nuclear cooperation agreements with France, Canada, UK, and Russia in respect to both power and desalination, and is developing its plans in line with the IAEA recommendations. It has signed a nuclear cooperation agreement with China, covering uranium mining in Jordan and nuclear power, and others with South Korea, Japan, Spain, Italy, Romania, Turkey and Argentina related to infrastructure including nuclear power and desalination.

In February 2015 Czech research and engineering firm UJV Rez and the JAEC signed a memorandum of understanding on the peaceful development of nuclear power. According to UJV, the MOU "assumes the strong involvement" of the Czech Republic in the construction of Jordan's first nuclear power plant. The agreement involves the design, construction and operation of nuclear plants and research reactors.

A January 2014 agreement with Saudi Arabia covers research related to nuclear energy and technologies, design, construction and operation of nuclear reactors. It includes "innovative new generations of nuclear reactors" plus safeguards technologies, nuclear materials controls, the preparation of nuclear safety and radiation protection legislation, environmental protection and human resources development. The agreement is also to "ensure transparency in the licensing and operation of nuclear facilities in the border area".

A full nuclear cooperation agreement with USA is pending, though the USA wants Jordan to emulate UAE and rule out uranium enrichment.

Jordan joined the Global Nuclear Energy Partnership (GNEP, now International Framework for Nuclear Energy Cooperation – IFNEC) in 2007.

Non-proliferation

Jordan has had a safeguards agreement in force with the IAEA since 1978, and an Additional Protocol in force since 1998.

Notes & references

General references

Jordan country profile, International Energy Agency